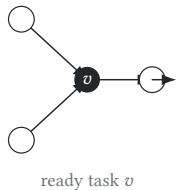


Task graph



CUDASTF scheduler

Candidate evaluator

GPU-ready $L(d)$ →

input-ready $A_v(d)$ →

compute p_v →

copy $K_v(d)$ →

d	F_v	K_v	S_v
b	min	$K_v(b)$	-
d^*	$F_v(d^*)$	lower	max
...	...	...	...

$$b = \arg \min_d F_v(d)$$

Prefix-safe admission

$$K_v(d) < K_v(b)$$

$$G_K(d) \geq \tau_v$$

$$d^* = \arg \max S_v(d)$$

$$a(v) = d^*$$

$$\text{if } P(d^*) \leq P(b) + \varepsilon_p$$

$$a(v) = b \text{ otherwise}$$

GPU queues

GPU 0



GPU 1



GPU 2



⋮

GPU 7



state update: $L(d)$, $M(x, d)$